

Deal & LP Origination Tool

Complete Product Flow Map & Business Rules

Version 3.0 | 15 July 2026 | All rules verified against production code | Supersedes v2.0 (8 July) | CONFIDENTIAL

What the product does

Three surfaces in one tool. **Deal engine**: sources UK/Ireland B2B software companies, qualifies them against the mandate, enriches with Companies House intelligence, finds and verifies founder contacts, scores fit, drafts and sends outreach, and tracks the pipeline through reply. **LP engine**: the same architecture pointed at investors — family offices, funds of funds, HNWI's — to invest through Averroes. **Intelligence Chat**: a database-grounded assistant that answers questions about any company or LP, never guesses, and runs a live web search only on explicit request.

What changed in v3

Area	Change
Mandate recalibration	Bands and size cap re-derived from the actual cheque: £15–40M equity for majority or significant-minority (25%+) stakes → investable revenue envelope £2.5–40M, core sweet spot £8–20M, hard cap £40M (was £50M). Applied retroactively to every stored row, including Qualified.
SmartFill order	The 3 hard filters now run FIRST as a cost gate. A failing company is marked Not a Fit with a stored reason (hover the badge) and consumes zero grounded-search budget — enrichment, Companies House and scoring are skipped entirely.
Contact waterfall	New 4-step, verification-gated contact engine: web personal → company's own site → retry search ladder → name-pattern inference. Every candidate must pass a mailbox check (Hunter) before it is stored; nothing verifies → the field stays blank. Applied retroactively.
Auto-draft	The moment a company qualifies, SmartFill drafts its outreach email (one ungrounded call, stored-signal news hook only). The button reads Review & Send when the card lands in the pipeline.
Lifecycle	New Engaged column between Qualified and Contacted: email sent, awaiting reply. Contacted is displayed as "Responded" — a reply exists. Manual advance skips Engaged; only a real send enters it.
Email sync	Scans Gmail All Mail (catches archived/filtered replies), matches replies by exact address, actual send recipient AND company domain, reads HTML-only emails, re-attempts stuck classifications, self-heals missed stage advances, and stamps events at the email's true time.
Intelligence Chat	Fourth feature: consultant-style answers strictly from the database; SmartFill nudge for unfilled companies; user-triggered grounded web search within the shared budget.
Test harness	The Internal Test row (source = 'Internal Test') is un-killable: recipient pinned to admin@, contact pinned, removal/lost auto-resets it to a fresh Qualified state for the next cycle.

Doctrine (CLAUDE.md, binding for all future work): BigQuery is the only owner of state — every page is a view of the same tables. A button with the same intent calls the same endpoint everywhere. Enrichment only replaces a description if the new one is longer. Timestamps record when the event happened, not when it was processed. All grounded AI goes through one weighted daily budget; paid grounding is structurally impossible.

1 · End-to-End Flow — Deal Side

From raw source to reply. All sources are real scrapers or uploads; no demo data exists anywhere.

Stage A — Getting companies in (NO AI, ever)

Source	What it does	Status on save
Excel/CSV upload	"pitchbook" in filename → full 152-column PitchBook parser; otherwise generic parser. Dedup on name; longer-wins description merge.	Uploaded
Conference scrapers	SaaStock (2022–25 archives), London Tech Week 2026 exhibitors + speakers.	Scraped
Network scrapers	Entrepreneur First alumni, Tech Nation portfolio.	Scraped
Ranking scraper	Startups 100 (startups.co.uk annual list).	Scraped
Directory scrapers	Category directories with address/category context.	Scraped

Stage B — The journey of one company

#	Step	What happens
1	Ingest	Row lands in Master Universe as Scraped/Uploaded. ingested_at written once, survives everything.
2	SmartFill	Hard-filter gate → contact waterfall → Companies House → fit score → revenue band → auto-draft (p.3).
3	Qualified	Card appears in the pipeline, outreach button already reads Review & Send. Sorted by fit score.
4	Send	One click after review. Recipient persisted; status → Engaged; contacted_at & outreach_sent_at stamped.
5	Reply	☒ Sync Emails logs the reply at its true received time, classifies it with AI, moves the card to Responded.
6	Human stages	Meeting → Due Diligence → Offer advanced manually (button or drag). Won/Lost close it out.

Manual override: clicking the status badge of any non-pipeline company (Not a Fit / Scraped / Uploaded / Under Review) offers "Qualify anyway" — human judgement beats the algorithm, and the override is logged with who did it.

Single source of truth: Universe and Pipeline render the same BigQuery rows filtered by status. SmartEnrich from either page updates the one record; every other page sees it on next load. Nothing is ever copied between pages.

2 · Inside SmartFill — The v3 Decision Flow

The hard filters moved to the FRONT: rejected companies now cost (almost) nothing.

Step	What happens	AI cost
0 · Caps	450 runs/day cap and the weighted grounding budget are checked before anything runs. Either exhausted → clean 429, no spend.	0
1 · Hard-filter GATE	Geography (UK/Ireland) + Industry (tech/SaaS) + Size (< £40M revenue, asset-aware AI when undisclosed). FAIL → status Not a Fit, reason stored in unfit_reason (hover the badge), local-proxy revenue band still computed, run logged at ZERO grounded weight. Nothing below this line executes.	1 ungrounded
2 · Contact waterfall	Web personal → own site → retry ladder → verified patterns; mailbox verification gates every candidate (p.7). Test row pinned to admin@.	1–2 grounded
3 · Companies House	Stringent name-gated match (p.5); accounts PDF parse (revenue, gross profit, profit, assets, cash, employees); registry intelligence: PSC ownership proof, charges/debt radar, SH01 capital events, accounts next-due.	1 parse call
4 · Fit score	5-metric model (p.6), only for Qualified companies. Bulk mode skips web-search scoring for Too Large rows.	0–2 grounded
5 · Revenue band	From filed → PitchBook → estimate cascade; Too Early / Target Band / Too Large stored on the row.	0
6 · Auto-draft	Freshly Qualified → outreach email drafted immediately (v7 structure, stored-signal hook only, never overwrites an existing or sent draft). Button shows Review & Send.	1 ungrounded
7 · Persist + audit	One UPDATE to the single row; description only replaced if longer; contact provenance and draft logged as activity notes.	0

SmartEnrich — the cheap refresh (already-filled companies)

Area	Rule
Contact email	ALWAYS re-verified against sources every run: confirmed (source noted), replaced (a source shows a different address), or CLEARED when no source anywhere publishes one — a stored address that cannot be sourced is treated as a legacy AI guess. Old value preserved in the activity note.
Name / LinkedIn / website	Gaps filled, never overwritten.
Registry intel	PSC, charges, capital events, accounts-due always refreshed (free CH calls).
Accounts PDF	Re-parsed ONLY if a newer filing exists.
Score	Re-scored only if previously unscored or new financials arrived.

Why the gate matters: in a 25-company bulk run over raw scraped data, historically most spend went to companies that were never going to qualify. Under v3 those companies consume one cheap ungrounded call each and stop.

3 · Qualification — The 3 Hard Filters & the Mandate Maths

All three must pass. Criteria live in BigQuery, editable in plain English via Criteria Chat.

Filter	Rule
1 · Geography	UK or Ireland. Fields: region, hq_country, hq_city, hq_location. AI infers when fields are sparse; keyword fallback without an API key.
2 · Industry	Technology / SaaS / tech-enabled. Checks sector, description, keywords, verticals, industry group. Traditional businesses without tech as core product do not count.
3 · Size	Revenue under £40M (v3, was £50M). Filed revenue when available; otherwise AI sizes from total assets (sector-aware), employees, funding, valuation. Buckets: Micro <£5M · Small £5–15M · Mid £15–40M qualify; Large >£40M rejected.

Where the numbers come from — the mandate derivation

Averroes invests **£15–40M of equity** into high-quality growing businesses in specialised markets, taking **majority or significant minority (25%+)** buy-out and growth positions. Working backwards:

Deal type	Cheque ÷ stake	Implied equity value
Majority (50–100%)	£15M/100% → £40M/50%	£15M – £80M
Significant minority (25–49%)	£15M/49% → £40M/25%	£30M – £160M

At 4–6x EV/revenue for specialised B2B software, the investable envelope is roughly **revenue £2.5–40M**, with the core sweet spot **£8–20M** (EV ~£40–90M, where both deal structures work comfortably). EBITDA cross-check at 8–12x: ~£3–15M.

Revenue bands (v3) — informational badge on every row

Band	Range	Meaning
Too Early	< £2.5M	Below the envelope; revisit later.
Target Band	£2.5M – £40M	Investable. Core sweet spot £8–20M scores 100 on revenue size; £20–40M tapers (needs a minority structure).
Too Large	> £40M	Beyond a £40M cheque even at a 25% stake. Also fails the hard size filter.

Retroactive application: a one-time, marker-guarded startup migration restamped the band on every stored row from its best revenue figure, and moved Qualified companies above £40M to Not a Fit with the reason on the badge. Companies already in live conversations (Responded and beyond) were left for human review.

4 · Companies House — Matching & Registry Intelligence

UK/Ireland companies only. A wrong company's accounts are worse than none.

Stringent matching — name similarity is a HARD GATE

Gate level	Condition	Confidence
Exact	Normalised names identical	HIGH
Exact core	Identical after stripping Ltd/Limited/PLC etc.	HIGH
Contains	One core name contains the other	MEDIUM
Fuzzy	High token overlap of distinctive words	MEDIUM
Partial	Weak overlap — accepted only with strong tie-breakers	LOW

Generic words (technology, software, app, solutions, ltd...) carry **zero** matching weight. Small tie-breakers (+active status, +SIC plausibility, +UK address, –dissolved) only separate candidates that already passed the gate — they can never rescue a bad name match.

What gets extracted once matched

Block	Fields
Accounts PDF (AI parse)	3 years revenue, gross profit (drives the margin-based revenue estimate), profit before tax, total/net assets, cash, employees, filing type. PDF stored in GCS and viewable from the company card.
Ownership (PSC)	Persons with significant control → one-line verdict, e.g. "Founder/family-owned (75–100% individual control)" — direct evidence for the founder-led thesis.
Debt radar (charges)	Outstanding charges count + summary. Charge holders reveal debt type: high-street bank vs venture debt vs invoice finance.
Capital events (SH01)	Last share allotment — funding rounds visible before any press.
Housekeeping	Status, incorporation date, SIC codes, accounts next-due (a company about to file is about to have fresh numbers).

SIC caveat: codes are self-reported and often lazy (62090 "Other IT"); useful for excluding holding/property vehicles, nearly useless for finding good SaaS. The tool treats SIC as a weak tie-breaker only. Overseas-registered entities (FC-prefixed) often lack SIC and PSC data entirely.

5 · Revenue Engine & Fit Score

Four estimation rules applied strictly in order; five scoring metrics; v3 curve.

Revenue estimation — the cascade never mixes tiers upward

Rule	Logic	Confidence
1 · Filed revenue	CH filed turnover first, then PitchBook revenue. Real numbers never blend with estimates.	HIGH
2 · Gross profit ÷ margin	GP from filings ÷ sector margin (software 80%, services 50%, unknown 70%). Overrides proxies; disagreement >2x is noted on the row.	MEDIUM
3 · Median of proxies	Employees x£100K · assets x sector multiplier (skipped for 64xx–68xx holding/property SICs; cash >60% of assets → assets–cash) · EBITDA x6.5 · valuation ÷6 · last round x3. Median of agreeing proxies.	MEDIUM/LOW
4 · Gemini blend	AI judgement joins only weak juries — never outvotes filings.	LOW

Averroes Fit Score — 5 metrics, composite = average of available (min 4 assessable)

Metric	Source	Judged on
Revenue growth	Local — CH filings	YoY filed revenue trajectory.
Employee growth	Grounded search	Hiring momentum, press signals.
Revenue size (v3 curve)	Estimation engine	<£2M: 0.1–0.25 · £2–5M rising · £5–8M: 0.7–1.0 · £8–20M: 1.0 · £20–40M: 1.0→0.5 · >£40M: 0.2
Business model fit	Grounded search	B2B SaaS-ness, recurring revenue, specialised market.
Market sentiment	Grounded search	Awards, funding news, analyst mentions — stored evidence later reused as the outreach news hook (free).

Cost note: bulk SmartFill (25/run) skips web-search scoring for Too Large companies — no budget burned scoring companies the mandate cannot buy.

6 · Contact Waterfall — Finding & Verifying the Founder Email (new in v3)

Four steps in strict order; a mailbox check gates every candidate; nothing verifiable → blank.

Step	Source	Rule
1	Personal email on the web	Grounded AI search: company site mentions, press, filings, articles. Aggregators (RocketReach, ContactOut...) are usable but their provenance is recorded — they mostly pattern-guess. Generic finds (info@) are demoted to step 2's tier.
2	The company's own website	Deterministic scrape of /contact, /about, /team, /legal + homepage: mailto links and address text. First-party personal beats everything; one generic (hello@) kept as fallback. Free, no AI.
3	Retry ladder	Only when steps 1–2 yield nothing usable: one sharper grounded search — the literal "@domain" string (addresses hide in PDFs and press footers), GitHub commits, conference speaker pages.
4	Pattern inference	jane.smith@ / jane@ / jsmith@... derived from the contact's name, mirroring any observed colleague pattern first. NEVER stored unverified — a guess must prove a mailbox exists.

Step 3 of every step — mailbox verification

Rule	Behaviour
Gate	Each candidate is checked against the mail server (Hunter email-verifier; GCP blocks direct SMTP). Deliverable → accepted, provenance + verification stored. Fail → next candidate, max 6 checks per company.
Nothing passes	The email field is left BLANK with the reason recorded. A wrong address is worse than no address.
Catch-all domains	Domains that accept all mail cannot confirm a mailbox. Strict by default (blank); VERIFIER_ACCEPT_CATCHALL=1 accepts found-in-source addresses on such domains, flagged as unconfirmable.
No verifier key	Degrades safely: best found-in-source candidate is stored and marked unverified; pattern guesses are never used. Adding HUNTER_API_KEY re-runs the retro pass in full-verification mode on the next deploy.
Retroactive	A marker-guarded startup pass re-resolves contacts for every already-SmartFilled company (no new AI calls) — updating, confirming or blanking, each change logged in the activity trail.

Provenance always: every stored email carries an activity note — where it was found and its verification status — so "where did this address come from?" is one click in the drawer, before any real email goes out.

7 · Deal Lifecycle & Outreach

The stage flow with Engaged and Responded, timestamps, and the v7 email engine.

Stage flow

Stage	Meaning	How it's entered
Qualified	Passed the 3 hard filters; outreach draft already prepared.	SmartFill, or manual "Qualify anyway".
Engaged	Outreach email sent, awaiting reply.	ONLY by actually sending — manual advance skips it.
Responded	A reply exists; dialogue open. (Stored as 'Contacted'; displayed as Responded.)	Auto on reply sync, or manual.
Meeting / DD / Offer	Human-driven stages.	Advance button or drag.
Won / Lost / Not a Fit	Closed outcomes. Not a Fit carries the reason on badge hover.	Manual, or the hard-filter gate.

Timeline rules: stage_entered_at resets on every move (drives kanban sorting and the >10-day ■ stale flag); per-stage first-entry stamps (qualified_at, contacted_at...) are written once and never overwritten; activity notes for email events carry the message's real Date header, not the sync time.

Outreach engine — email structure v7 (fixed order, wording varies)

Hi {first} → pleasantry → **who we are** (Partner at Averroes, proven product, next stage of growth) → **follow + conviction**, strictly two sentences from the investor's seat ("We have been following X and really like <specific>. We believe <problem> is a real pain point and the potential is huge.") → portfolio proof (**Journey and Glowday**, hands-on operational support alongside capital) → humility → "open to a quick 20-minute call in the next couple of weeks?" → "Look forward to hearing back from you." → "Best," (signature auto-appended: Maria Beatrice Carrara, Partner).

Guardrail	Rule
Never invent	No facts, numbers or news beyond stored data + the verified signal. Never quote their financials back at them.
Natural name	AI strips legal suffixes and registry casing: "FIDO TECH LTD" is written "Fido Tech"; genuine brand styling kept.
Style bans	Zero em dashes anywhere. No jargon (leverage, ecosystem, synergies...), no "I hope this email finds you well", no exclamation marks. 120–160 words rich / 90–115 thin.
News hook	Stored scoring signal first (free); a live search only when drafting manually and budget allows. Auto-drafts never search.
3-state button	Outreach → Review & Send (saved draft opens, no regeneration) → Email Sent ✓. Same shared component on Universe and Pipeline; drafting never changes the stage — only a successful send does.

8 · Email Sync — Reply Intelligence

One button reads Beatrice's mailbox and keeps the pipeline honest. Heavily hardened in v3.

Rule	Behaviour
Scan	Gmail All Mail via IMAP (read-only): last 30 days, read or unread, archived or filtered — up to 500 most recent. Direction detected per message (from beatrice@ = sent).
Matching (3 levels)	Exact contact email → actual send recipient (persisted at send time) → COMPANY DOMAIN: a reply from anyone @company-domain matches the company (founders often reply personally when we wrote to hello@). Free-mail domains excluded; an address on both a company and an LP resolves to the company.
Process once	Dedup by Message-ID: each email is logged, classified and stamped exactly once, at its true received time. Re-syncing costs nothing.
Classification	Gemini (≤25/run): interested / not_now / declined / question / other. HTML-only emails are converted to text first. Replies stuck as "unclassified" are re-attempted on later syncs within the same budget.
Stage effect	Any reply advances Engaged → Responded (your chosen rule: you judge tone from the 📧 badge). A self-healing pass also advances companies whose reply was logged but whose stage never moved.
Test harness	The Internal Test row bypasses send guards, advances from any pre-Responded state, keeps its contact pinned to admin@, and auto-resets to a fresh Qualified on any removal — an always-available end-to-end test loop.

Event truth: a founder who replied Tuesday night shows Tuesday night in the activity timeline, regardless of when Sync was pressed. The email_log table is the permanent record of every exchange with known contacts.

9 · LP Module & Intelligence Chat

The second origination engine, and the fourth surface added in v3.

Investor (LP) module — family offices, FoFs, HNWIs/UHNWIs

Piece	What it does
Sources	Mine from high-fit companies' cap tables · PitchBook LP export upload (152-column parser, source = filename) · Praxis Rock directories · CH registry search (name patterns + SIC 64303/66300).
InvestorFill	1 grounded call: 4 fit criteria (UK/Europe/KSA geography, PE appetite, £250K–5M ticket, tech affinity; 3-of-4 rule) + CH enrichment for UK entities (PSC principals, officers, net assets as AUM proxy).
Workflow	Stages Identified → Researched → Contacted → Meeting → Committed/Passed; LP outreach draft + send; CSV export; chronological table.

Intelligence Chat — database-grounded Q&A; (new in v3)

Rule	Behaviour
Data-only	Answers strictly from the companies + LP tables: full records for entities named in the question (typo- and spacing-tolerant matching), plus a full-universe index for aggregate questions ("Target Band above 70 fit?").
Never guesses	Outside knowledge is forbidden even for recognisable names. If the database lacks the answer, it says so — and offers a button.
Web search on request	The  button runs ONE grounded search (weight 1 in the shared budget), and the answer is visibly tagged as coming from the live web.
SmartFill nudge	Asking about an unfilled company gets the honest answer plus "run SmartFill for contacts, financials and the fit score".
Format	Consultant-style: one takeaway line, small bold-headed sections, bullets. Simple questions get a single paragraph.

10 · Cost Guards & Scenario Playouts

The zero-paid-grounding guarantee, and realistic cases traced through the v3 rules.

Cost architecture — paid grounding is structurally impossible

Guard	Value	Mechanism
Run cap	450/day	SmartFill + SmartEnrich runs (gated runs included), counted from the activity log; resets midnight UTC.
Grounding budget	1,400/day	Weighted worst-case: SmartFill x4, SmartEnrich x3, InvestorFill x1, news/chat search x1 — 100-call buffer under Google's 1,500 free tier. Every grounded call is enforced BEFORE it runs and logged after.
Bulk batch	25/run	Bulk SmartFill processes at most 25 eligible companies per click, with a pre-run funnel estimate.
No AI at ingest	0	Scrapers and uploads never touch AI. SmartFill / InvestorFill / Chat are the only AI layers.
Startup migrations	once	Marker-guarded (bands v3, contacts retro) — idempotent, never repeated, logged.

Scenario playouts

Scenario	How v3 handles it
S1 · PitchBook upload, London B2B SaaS, £12M filed revenue	Gate: geo ✓ tech ✓ size Small/Mid ✓ → contact waterfall finds jane@ on the team page, verified deliverable → CH exact-core match: 3yrs accounts + "Founder-owned (75–100%)" PSC → score ~high (revenue size 1.0 in the core band) → Target Band → auto-draft ready → Qualified with Review & Send.
S2 · Scraped conference exhibitor, US-based	Gate fails geography on the single ungrounded call → Not a Fit, reason on badge hover, zero grounded spend. Reversible any time via "Qualify anyway".
S3 · £45M-revenue scale-up, previously Qualified under v2	v3 migration: band → Too Large, status → Not a Fit ("Revenue £45M exceeds the £40M cap"), activity logged. If it were already in Responded, it would be left for human review.
S4 · Sent to hello@, founder replies from nelson@same-domain, email was archived by a filter	All Mail scan finds it; domain-level matching resolves it to the company; classified; ☒ badge; Engaged → Responded; activity note at the reply's true time.
S5 · No email anywhere for a qualified founder	Waterfall exhausts web → site → retry → patterns; verifier confirms nothing → field stays BLANK with the reason stored. The draft is still ready — type a recipient at send time.

Document ends. Version 3.0 — all rules verified against production code on 15 July 2026.